

NFINGER: Personalized Backside Gesture Recognition on Smartphones Using Back-mounted NFC Tag Array

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Abstract

Touch gestures are a primary input method for smart devices, yet touchscreen interfaces are limited by finger occlusion and constrained screen space. To address these issues, we present NFINGER, a personalized gesture sensing system using a passive array of commercial NFC tags placed on the back of a smartphone. NFINGER is low-cost, easy to deploy, and operates by sensing changes in amplitude and phase of backscattered signals caused by bio-impedance coupling during finger contact. These signal variations depend on both gesture type and the user's unique finger impedance, enabling NFINGER to simultaneously recognize gestures and identify users. To improve sensing robustness, we introduce a T-shaped tag layout that reduces inter-tag interference, along with signal processing techniques such as dynamic range expansion and adaptive normalization to enhance signal quality. We evaluate NFINGER through extensive experiments involving multiple users and gestures. Results show an average recognition accuracy of 95.7% across eight gesture types and 94.3% for user identification, demonstrating the system's potential for seamless, user-aware interaction in ubiquitous computing environments.

CCS Concepts

• Human-centered computing → Ubiquitous and mobile computing.

Keywords

Gesture Recognition, NFC Tag Array, Wireless Sensing

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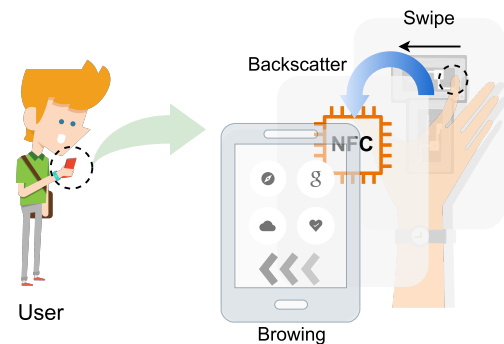


Figure 1: Interactive process of NFINGER.

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1 Introduction

Touch gesture recognition is a basic HCI mechanism for smart devices, especially smartphones and tablets. By performing simple operations such as tapping or swiping on the touchscreen, users can easily control their devices[17]. Although this interaction method is popular, the high cost of touchscreen hardware restricts gesture input primarily to the front surface of mobile devices, which imposes limitations on the user experience. For example, during the interaction process, the user's finger often blocks the content displayed on the touch screen. Moreover, smartphones are commonly used one-handed with only the thumb touching the display, and it is often difficult for the thumb to reach the entire screen.

While some of the latest mobile devices have introduced new interaction methods, such as sliding gestures and eye-flipping[1, 12], these methods have higher requirements on the relative position of the device held by the user, and the recognition accuracy and response speed still fail to meet the actual needs in daily scenes. In contrast, leveraging the back of the device as an additional input surface for touch gestures not only avoids content occlusion[11, 13], but also utilizes the free index finger to operate independently, thus expanding the input space of the mobile device and providing a brand new interactive experience to the user.

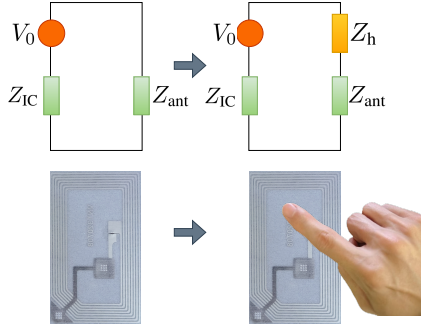


Figure 2: Equivalent circuit of a tag with the fingertip touching.

In gesture recognition research, RFID has attracted significant attention due to its unique backscattering properties[2, 3, 6, 8]. Inspired by this, we introduced NFC technology with similar working mechanism and designed NFINGER, a touch gesture recognition system for back-of-device interaction on a smartphone. As shown in Fig. 1, using an array of commercial NFC tags attached to the back of a smartphone, NFINGER can accurately sense the user’s touch behavior by analyzing the physical-layer signals collected during the communication between the NFC reader and the tag. Specifically, when a human touches the NFC tag at different locations, it causes noticeable changes in the amplitude and phase of the reflected signal from the tag[16], resulting in distinct patterns. The system realizes the recognition of touch gestures by extracting and processing features from these reflected signals. In addition, the reflected signals from the tags also contain biometric differences associated with the individual[5], which NFINGER can use to accurately identify the user. Through this mechanism, we can establish a privilege control policy that enables backside interaction only for authenticated users and assigns control privileges based on the user’s identity.

The contributions of this paper are summarized as follows:

- We designed a personalized gesture recognition system called NFINGER, which utilizes the built-in NFC function of smartphones and low-cost commercial NFC tags.
- By performing fine-grained modeling of the time-domain and amplitude-phase features of NFC reflection signals, NFINGER enables both touch gesture recognition and user identification.
- We constructed an experimental dataset containing 10 users, covering a wide range of environmental conditions. The experimental results show that NFINGER achieves 95.74% accuracy in touch gesture recognition and 93.11% accuracy in user identity recognition.

2 Feasibility of NFINGER

2.1 Introducing Human Impedance

The human body can be modeled as a complex impedance network consisting of resistors and capacitors [5, 10]. As shown in Fig. 2, when a fingertip touches the NTAG213 tag, it introduces an additional impedance Z_h (300–1000 Ω) into the circuit, which alters the antenna impedance Z_{ant} . At this point, the equivalent circuit of

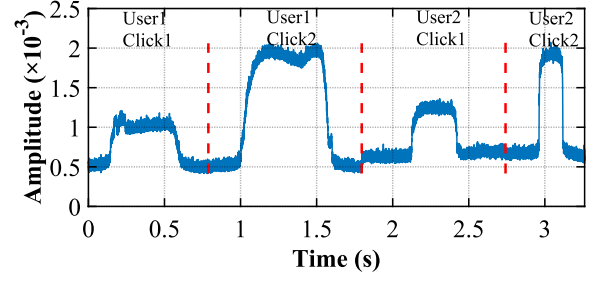


Figure 3: Amplitude changes caused by gestures of different users.

the fingertip touching the tag consists of an induced voltage source V_0 , chip impedance Z_{ic} , antenna impedance Z_{ant} , and additional impedance Z_h .

This impedance perturbation modifies the electromagnetic coupling characteristics between the tag and reader [3, 20], reflecting in changes to the reflection coefficient. The reflection coefficient Γ is defined as:

$$\Gamma = \frac{Z_L - Z_S}{Z_L + Z_S}, \quad (1)$$

where $Z_S = Z_{ant}$ and $Z_L = Z_{ic}$. Therefore, the impedance Z_h introduced by finger contact alters Z_{ant} , resulting in measurable changes in the amplitude and phase of the backscattered signal.

2.2 Discriminative Features

We investigated the impact of figure touch on NFC tag reflection signal. As shown in Fig. 3, we observed that touching the left and right sides of the same tag surface with the finger produces reflected signals with significantly different amplitudes, due to varying coupling conditions[10].

Furthermore, we observed that signal perturbations differ across users, which is mainly attributed to differences in skin conductivity and tissue composition, exhibiting unique bioimpedance characteristics[7]. As a result, these spatial and individual differences enable us to extract features from the backscattered signal for both touch gesture recognition and user identification.

3 System Design

3.1 Tag Array Design

Due to the limited space on the back of smartphones, multiple NFC tags often operate within the same magnetic field. However, in dense configurations, mutual inductive coupling between tag antennas can alter their resonant frequencies, leading to signal interference and reduced recognition accuracy[8]. For instance, when tags are placed side by side, strong coupling causes noticeable frequency shifts that degrade signal stability[9]; In loop-shaped layouts, some tags are positioned farther from the reader antenna, resulting in weaker backscatter signals due to low induced voltage and unfavorable orientation[15].

To mitigate this issue, we adopt a T-shaped layout. By arranging tags at right angles, the design achieves orthogonal coupling, which minimizes resonance shifts caused by mutual interference. Although this layout introduces edge flux leakage, the resulting

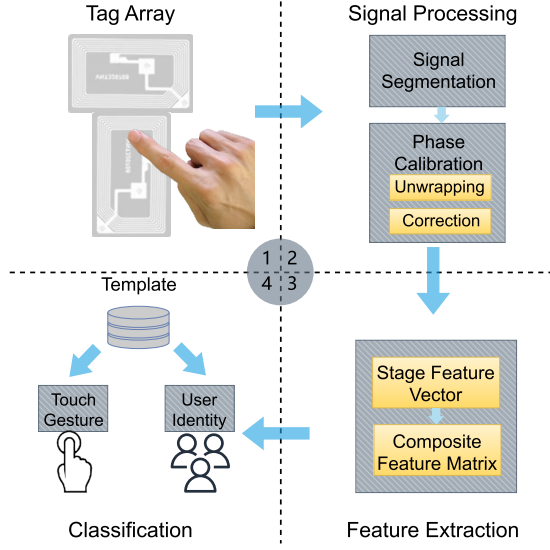


Figure 4: System architecture of NFINGER.

phase noise can be corrected through calibration during signal processing. Additionally, the unique phase characteristics generated by this orthogonal structure help distinguish signals from each tag. The mixed orientation also increases directional diversity, enabling richer gesture recognition without sacrificing signal quality.

3.2 Signal Processing

3.2.1 Signal Segmentation. To extract the signal segments corresponding to touch events from the received backscatter, we improved traditional peak detection methods[4] and introduced an indirect segmentation strategy based on background modeling. This method first identifies static regions (i.e., intervals without touch events) and then infers the boundaries of touch events to locate non-static touch events. As illustrated in Fig. 5, the procedure is as follows:

We first construct a background model using the initial stationary segment of the signal. From this segment, we compute the mean μ_{bg} , variance σ_{bg}^2 , and range R_{bg} as reference statistics.

Next, we use a sliding window to traverse the signal in real time. For each window, we calculate the local mean μ_w , variance σ_w^2 , and range R_w . If a window satisfies both $\sigma_w^2 \leq \sigma_{bg}^2 + 0.8 \sigma_{bg}^2$ and $R_w \leq R_{bg} + 0.6 R_{bg}$, it is classified as a static window. By marking all stationary windows, we effectively segment the touch induced signal bands for gesture recognition.

3.2.2 Phase Calibration. When the phase change triggered by touching the tag crosses this boundary, the true phase may be incorrectly recorded as jumping from 2π to 0 (positive wrap) or from 0 to 2π (negative wrap)[18, 19]. We use phase unwrapping to eliminate these errors and restore a continuous absolute phase sequence.

To remove the phase noise introduced by the T-shaped layout, we first measure the reference phase of an isolated single tag in an interference free environment. Then, we construct a coupling matrix from the measured T-shaped data and employ the least

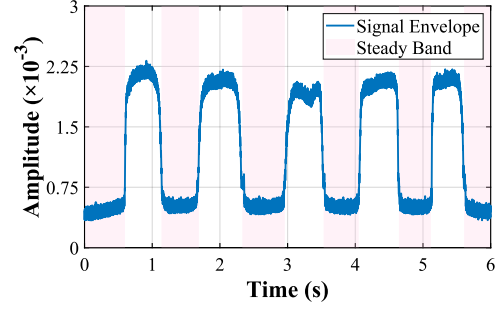


Figure 5: Stationary band detection.

squares method to solve for the optimal correction term. Finally, applying this correction term completely eliminates the phase noise in the T-shaped layout.

3.3 Feature Extraction

After signal processing, we obtain the signal segments corresponding to touch gestures and select four features of the backscatter signal that reflect touch behavior: mean amplitude, amplitude slope, duration, and mean phase. To enhance temporal representation, we divided the time domain signal of each gesture four key phases-*Initialization, Execution, Holding, Release*. For each phase, we construct a corresponding specific feature vector:

$$\mathbf{f}_i = [A_{\text{mean}}^{(i)}, S_{\text{mean}}^{(i)}, T_{\text{mean}}^{(i)}, \phi_{\text{mean}}^{(i)}] \quad (2)$$

Finally, the four phase vectors are combined into a composite feature matrix of size 4×4 :

$$\mathbf{M} = \begin{bmatrix} A_{\text{mean}}^{(1)} & A_{\text{mean}}^{(2)} & A_{\text{mean}}^{(3)} & A_{\text{mean}}^{(4)} \\ S_{\text{mean}}^{(1)} & S_{\text{mean}}^{(2)} & S_{\text{mean}}^{(3)} & S_{\text{mean}}^{(4)} \\ T_{\text{mean}}^{(1)} & T_{\text{mean}}^{(2)} & T_{\text{mean}}^{(3)} & T_{\text{mean}}^{(4)} \\ \phi_{\text{mean}}^{(1)} & \phi_{\text{mean}}^{(2)} & \phi_{\text{mean}}^{(3)} & \phi_{\text{mean}}^{(4)} \end{bmatrix} \quad (3)$$

The matrix $\mathbf{M}$ captures how amplitude and phase vary across each touch phase. By designing a classifier to analyze $\mathbf{M}$, we can accurately distinguish different touch gestures and identify users.

3.4 Gesture-User Classification

After extracting feature vectors for touch gestures, we use a customized CNN+Bi-LSTM+Attention model architecture[14] to infer both the touch gesture and user identification. Its core idea is single-network multi-task learning, combining hierarchical feature extraction with two-stage progressive training. In this architecture, the CNNs extract spatial information, the Bi-LSTMs capture forward and backward temporal dependencies of those amplitude features, and the attention mechanism dynamically focuses on key action segments. The shared high-level semantic representation is finally branched into two task-specific heads for touch gesture recognition and user identity classification.

To address the problem of extremely weak backscatter signals, we add a data augmentation module before the CNN layer that applies dynamic range expansion to amplify subtle features:

$$x'_t = \text{sign}(x_t) \frac{\log(1 + \lambda)}{\log(1 + \lambda |x_t|/\sigma)}. \quad (4)$$

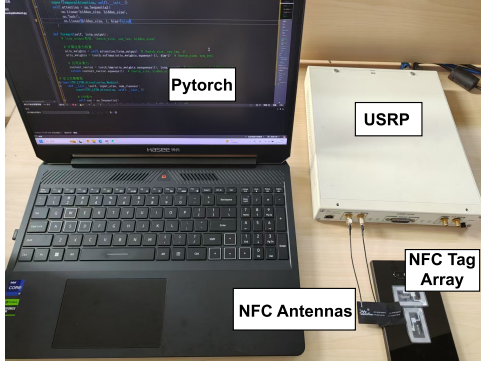


Figure 6: Experimental environment.

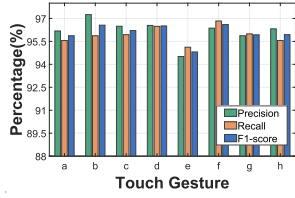


Figure 7: Precision, recall and F1-score of classification.

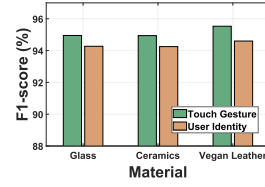


Figure 8: F1-score of classification on different materials.

This architecture balances computational efficiency and classification accuracy through multi-task parameter sharing and the dynamic attention mechanism.

4 Evaluation

4.1 Experimental Setup

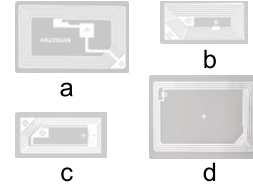
The NFINGER prototype uses a USRP to emulate an NFC reader[15], with two independent ports connected to the NFC transmitter and receiver antennas, respectively. Communication between the USRP and tags is implemented by GNU Radio on a laptop, and the received signals are processed and used for model training in Pytorch 1.10.1, as shown in Fig. 6.

In our experiments, 10 volunteers are employed as human subjects. We consider 8 commonly used touch gestures, including tapping and swiping on both sides of the horizontal and vertical tags. These gestures are sufficient to cover the requirements of typical daily interactions. Each user repeats every gesture 20 times, yielding 160 samples per person.

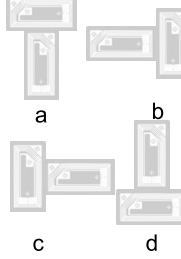
4.2 System Performance

4.2.1 Overall performance. We evaluated NFINGER’s overall performance across all volunteers and experimental conditions. Fig. 7 shows the F1-scores for eight touch gestures, and the overall gesture classification accuracy is 95.74%.

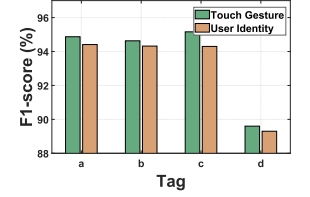
4.2.2 Material. We evaluated classification performance of smartphone back covers made from three different materials. Fig. 8 shows the F1-scores for gesture recognition and user identification tasks.



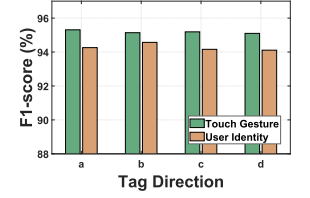
(a) Different tags.



(c) Tag direction.



(b) F1-score of different tags.



(d) F1-score of different tag directions.

Figure 9: System performance analysis on different conditions

The results indicate that typical back cover materials have minimal impact on system accuracy—their similar capacitive properties leave the finger-to-tag coupling largely unaffected.

4.2.3 Tag Structure. To evaluate the impact of different NFC tag geometries on performance, we repeated gesture recognition and user authentication tests on four tags with different geometries. Fig. 9b shows these results, indicating that tags with similar coil designs achieve comparable accuracy. However, Type-D tags have a larger surface area, which reduces effective contact between the fingertip and the circuit, resulting in weak signal interference and degraded performance.

4.2.4 Tag Direction. To explore the effect of tag orientation, we rotated the T-shaped array through four distinct directions. As shown in Fig. 9d, although the orientation alters the propagation path of the signal, all configurations remained within 1 cm of the reader antenna, with the maximum F1 score variation of only 0.4%.

5 Conclusion

In this paper, we introduced NFINGER, a low-cost, easily deployable gesture sensing system that leverages passive NFC tags on the back of smartphones to achieve personalized touch gesture recognition and user authentication. The design of NFINGER is based on two key observations: user touch introduces changes in body impedance, which in turn affects the characteristics of the backscattered signal; and body impedance is unique across individuals. Additionally, we address the technical challenges involved in developing a practical system. Once mobile devices fully support NFC access, our solution can be directly deployed on smartphones.

Acknowledgments

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